LIGHTYEAR

MS91 Declared dark factory operations run

Lightyear | Milestone report | 27 September 2026

Word edition

Outcome and customer value

MS91 completed a fresh agent-generated iDempiere journey on native Oracle and PostgreSQL under a judge, controller, comparison register and budget declared before generation. The first candidate passed the combined deterministic judge. Full database captures subsequently reproduced the passing result offline.

Customer claim: One bounded operations journey passed with zero human-authored candidate or repair-message bytes and zero controller changes during the campaign. The result satisfies the declared conditions for a bounded dark factory run. It does not establish full application, schema or platform qualification.

Campaign measure	Recorded result
Combined native gate	Passed on the first candidate
Approved and consumed calls	11 approved; one builder call used
Analyst calls and repairs	Zero; repair loop not exercised
Human-authored repair bytes	Zero, provenance verified
Controller changes and interventions	Zero
Unresolved row and trace differences	Zero and zero
Total campaign time	745.157 seconds, or 12 minutes 25 seconds
Cleanup	Complete; no owned containers, networks or volumes remained

Why this milestone matters

MS89 demonstrated a bounded repair workflow but required disclosed controller revisions. MS90 applied a separately authorized comparison policy to retained evidence while preserving the original failed verdict. MS91 moved the acceptance decision before generation: the factory knew its permitted task, and the judge already knew the rules by which the result would be accepted.

The factory generated the Java candidate. The engine executed and judged the same candidate on both databases. A person had already approved the comparison scope and resource budget. Nobody changed expected business values, the judge, the controller or the budget during this run.

The observed journey

The generated application journey created a customer and stocked product, used fractional prices, tax, a discount, Unicode text, an empty optional field and a fractional shipment timestamp, and completed two partial shipments followed by one aggregate invoice and payment. It then created and reversed a credit note, exercised two overlapping explicit-row-lock updates, and injected a pre-commit failure followed by rollback and caller-keyed retry.

These are several operations within one bounded journey, not a claim that every credit, reversal, concurrency or recovery scenario has been covered.

The declared comparison policy

Numeric eligibility follows native column types

The version-3 judge uses the verified datatype inventory, which identifies 9,393 Oracle NUMBER to PostgreSQL numeric columns. Native catalogs must match the frozen declarations. Numeric trace values must also bind to independently captured cells; the stock aggregate binds to captured numeric stock rows. A field name or numeric-looking text is insufficient.

Equality uses exact decimal values without rounding or tolerance. Exponent notation, nonfinite values, floating-point inputs and unsupported wrappers are rejected, including when both sides contain the same invalid spelling. SQL nulls remain nulls. An unbound intermediate trace is compared as an exact string rather than receiving a numeric exception.

The initial work order specifies consistent plain-decimal trace serialization. That requirement was fixed before generation. Earlier retained observations still show their original formatting differences; they were not rewritten to satisfy this campaign.

One register for rules and retained differences

The frozen register contains 15 column rules, one engine-identity metadata rule and 6,024 historical difference witnesses. Every rule has an owner, effective date, review date and measured scope. Numeric eligibility spans the complete typed inventory, while UUIDs, audit clocks, workflow text and engine metadata retain their separate, bounded predicates.

Rule or observation	Required treatment
Numeric cell values	Exact decimal equality with captured provenance
Numeric audit clocks	Differences need their separate owned clock rule
UUIDs and audit metadata	Re-execute the registered validation predicate
Unknown or unmeasured rule scope	Refuse admission rather than substitute zero
Unbound intermediate trace values	Exact string comparison

The accepted timestamp decision

The existing signed decision covers only `adempiere.m_inout.shipdate`: Oracle DATE compared with PostgreSQL timestamp without time zone in verified UTC capture sessions. Oracle must have no fraction, and every date and time component through seconds must match. The rule does not round into a different second. Raw fractions and their loss remain visible.

Owner: Howard Weale. **Effective:** 26 September 2026. **Review:** 25 December 2026. Another column, another datatype, an unsupported timezone context, or a business requirement for subsecond ordering or timing remains outside the decision.

What the judge observed

The new run validated 2,746,888 numeric cell observations and retained 6,038 admitted raw row differences plus 15 admitted trace differences. No row or trace differences remained unresolved. These are observation and witness counts, not millions of independent test cases; much of the observed database content is inherited seed data.

Frozen execution and measured effort

Declare once and execute once

The plan pins the implementation and public-input hashes, comparison register, database images, application revision, CLI binary identity, requested model and reasoning setting. It authorizes at most

11 total client calls and four hours of local execution. Each client call has a 15-minute limit. The candidate is limited to one Java file, 60,000 bytes and 650 changed lines.

There is no resume or amendment command for this campaign. A changed judge, controller, input or exhausted budget stops it. Further work requires a new declaration and authorization. Every attempt binds its judge hashes and rule version to the generated candidate and native result.

The analyst channel permits only compiler diagnostics, supported API type mismatches and a verified out-of-footprint table and call. It can select closed diagnostic records, not send free-form repair instructions or expected business outputs. In this run, no analyst call was needed.

Complete recorded cost

Measurement	Observed value
Client calls	One builder, zero analysts
Input and output tokens	17,321 and 11,441
Cached input tokens	Zero
Failed calls and native attempts	Zero and zero
Paired native attempts	One, covering both engines
Generation time	349.641 seconds
Native execution and gate time	387.656 seconds
Total campaign time	745.157 seconds, including controller overhead
Call accounting	Complete for the observed client invocation
Dollar billing	Unknown under the signed-in transport
Cloud resources started	None; local Docker execution only

The requested model was `gpt-6-astra` with reasoning `high`, through Codex CLI `0.155.0-alpha.9.2`. The executable version and SHA-256 were pinned. The provider-resolved model snapshot and internal provider retries are not exposed. Organization-controlled API migration remains deferred; unavailable dollar cost is not reported as zero. Local electricity was not metered.

Meaning of zero repair bytes

The audit matches candidate bytes to the model proposal and verifies the input/result chain. Zero repair bytes excludes preparation of the work order, controller, judge and policy. This first-candidate pass does not demonstrate analyst-led repair success.

Builder input provenance

Inputs inherited MS86's API example and collection-query lock lesson; MS88/MS89's Boolean-accessor, accounting-signature and no-reposting lessons; and MS89/MS90's decimal-format lesson. None changed during MS91. "First try" means the first candidate using accumulated lessons. The input provenance register identifies their origins.

Published evidence and remaining limits

Complete offline replay

The MS91 publication contains 7,509 files in a 61,948,306-byte archive: full captures and raw rows, catalogs, the journal, generated candidate, prompts, model responses, call accounting, frozen implementation, rules and receipts. Offline verification checked the signed identities, reconstructed the

artifacts, audited provenance and reran the complete native gate. It reproduced the pass without another database execution or model call.

The older MS90 full-capture publication is also included. Its 7,379 files reproduce both the original failing gate and the separately passing version-2 assessment. MS91 does not replace either historical verdict.

From the repository root, with dependencies installed:

```
$env:PYTHONPATH='src;.'
$key='c2fbae3453e8bcae03e0830a9eb83f310f027ad51929deee967ecee1a41f399f'
python -m tools.publish_frozen_campaign verify `
  --output docs/calibration/idempiere-declared-factory `
  --trusted-key-sha256 $key
```

Expected status: `verified-complete-frozen-campaign`, judge `idempiere-declared-comparison-v3`, one replayed passing gate, and zero new model calls or native executions. Establish trust in the operator key separately. Local signatures provide attribution and tamper evidence, not independent attestation.

Evidence identity

Frozen campaign plan SHA-256:

`cc547e958c97bd6b500fb57c5bfce9fa21824453fbf55178465ab628a0df8a3d`

Signed campaign receipt SHA-256:

`e121c5c9fa3d789054651f8f37ea7b64c3e927a4697a18e39d2ce367f9ec35fe`

Native run: `journey-d8a88a4c337845d4bcaf42ef976ff028`. Every attempt's judge hashes and the rule-register identity are retained in the signed result and archive.

Limits and next work

Committed outcomes have independent full-database readback. Intermediate lock waits and rollback checkpoints still include application-trace assertions. This is not independent transient-event monitoring, crash or restart recovery, sustained load testing, or full application, schema or platform qualification. The receipt keeps those broader qualifications false.

The next useful evidence is a separately declared run with independent lock and rollback observation, followed by a controlled repair exercise under another frozen budget. The timestamp owner must review the precision decision by 25 December 2026. None of that future coverage is implied by the current pass.

Validation and references

Twenty-one new adversarial and lifecycle tests and 72 existing journey tests passed before generation. The fresh native campaign and complete offline replay then passed. Independent Docker inventory checks confirmed cleanup after the run.

- MS91 evidence and replay guide
- Signed campaign result
- Frozen plan
- Owned comparison register
- Complete MS90 captures